

Naked came the crook

A nude man who came in from the cold to rob a gas station early yesterday had a decided advantage, police said. The attendant was female. She fled, leaving the unarmed man free to empty the cash register, then stroll casually from the scene of the crime.

FREEBASING COCAINE

by Dr. Atomic

Freebasing cocaine is basically a California phenomenon, but it's a practice that is popular with entertainers and with others who can afford to indulge in the pleasures of cocaine. Freebase cocaine is smoked in a special, glass water pipe called a freebase pipe, and after taking a toke the onset of the high is quick: it comes on faster than snorting and almost as quick as an i.v. injection -- it's like injecting cocaine without using a needle. After inhaling the freebase cocaine vapors, your hearing drops, and you get an incredible rush even before enough time passes to exhale the smoke. Unfortunately, the rush and the high don't last long, and the desire to smoke some more coke is compulsive. In fact, it is so compulsive that people who hang around the freebase pipe, impatiently waiting to get another toke, are known in the vernacular as "freebase vultures". But before the cocaine can be smoked, it must first be prepared.

The cocaine purchased on the street is usually cocaine hydrochloride (HCl), a water soluble salt of cocaine that is suitable for snorting or injecting, but not for smoking. Cocaine HCl burns at a high temperature, about 200°C, and if it's smoked, much of the cocaine gets carbonized, burned up, instead of reaching your lungs as vapors. But, by changing the cocaine HCl to cocaine freebase, you get more of the desired cocaine vapors and less carbon because the freebase vaporizes at a much lower temperature than the cocaine HCl does.

All it takes to change the cocaine HCl into cocaine freebase is a little home chemistry. It's easy: if you can bake brownies by following a cook book, you can freebase coke. The only supplies needed are some inexpensive chemicals and equipment that are easily obtainable at your local paraphernalia shop.

Equipment and Supplies

- 1 freebase water pipe, glass
- 2 screens, fine mesh, for pipe
- 1 glass freebase vial, 1 oz, with top¹
- 1 mirror
- 1 single edge razor blade
- 1 box baking soda
- 1 bottle of petroleum ether or ethyl ether²
- 1 book matches or butane lighter

NOTE 1: Ethyl ether and petroleum ether will dissolve many plastics, so the tops of freebase vials are specially made of ether resistant plastic.

NOTE 2: Use caution when handling ether. The vapors of both ethyl ether and petroleum ether will ignite explosively near an open flame. Make sure that the room is well ventilated when extracting with ether. When freebasing in the kitchen, make sure the pilot lights are out on the stove and the hot water heater if they are nearby. Also, don't smoke or light matches while there are still fumes in the air.

The Freebase Process

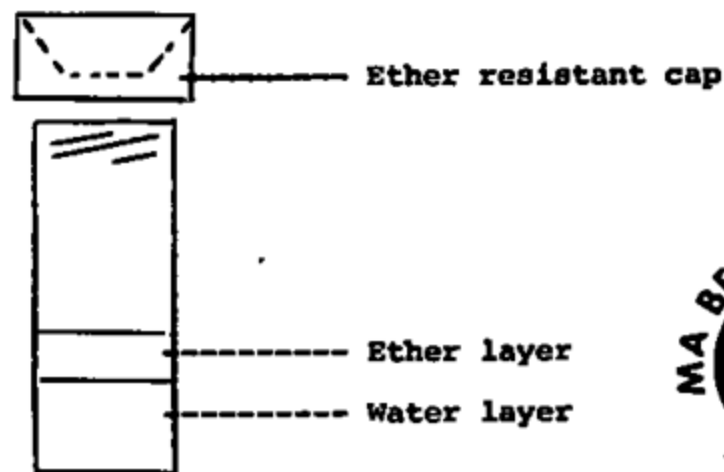
- 1) To a 1 oz glass freebase vial, add 4ml to 6ml of warm water. Less than 1/4 of the vial is more than sufficient water.
- 2) Dissolve 1/4 to 1/2 gram of cocaine HCl in the water to make a cocaine solution. Shake or stir if necessary to dissolve the cocaine.
- 3) Add about 1/4 gram, more or less, of baking soda to the cocaine HCl solution. It is better to have an excess of baking soda than not enough. Next, shake well. This changes the cocaine HCl to the freebase.
- 4) Using a glass eyedropper, add 2ml to 3ml of ether. Shake well. The ether extracts the freebase cocaine from the water layer. As a rule of thumb, use half as much ether as water. Since ether and water do not form a solution, the ether will rise to the top and form a distinct layer.



TAP

Room 603
147 W. 42 St.
New York 10036

No. 70



Because the cocaine freebase is more soluble in ether than in water, the ether layer will contain most of the freebase; in effect, the ether has extracted the freebase cocaine from the water layer. This first ether extraction is known as the "first wash". The water layer can be washed one or two more times with ether to extract the small amount of freebase remaining after the first wash.

5) Siphon off the ether layer with the eyedropper, making sure not to take any of the water layer. Drop the freebase saturated ether carefully onto a clean mirror or glass surface. When the ether evaporates, a white powder should remain: this is the cocaine freebase, and it's ready to smoke. So what are you waiting for?

The freebasing process removes some of the water soluble contaminants (cuts) like mannitol and lactose, so the yield, i.e. the weight of the cocaine freebase obtained will weigh less than the cut-coke that was started with; however, no significant amount of cocaine is lost; only the cut is removed. Thus, a gram of cocaine HCl that is only 25% pure is not a gram of cocaine but a 1/4 gram of cocaine, and the yield of freebase cocaine, for this particular sample, will be slightly less than 1/4 gram.

The cocaine freebase, however, is nearly pure, compared to the starting material, and a smaller dose of the freebase will be just as potent as a larger amount of the cut cocaine. So, start with a small hit, a match size line or less (20mg to 50mg). Remember, just like snorting or injecting, you can consume too much by smoking. Be careful how much you smoke, and be careful, too, for police and informers: cocaine is still illegal. Have fun with your chemistry projects, stay high, and stay free.

Man nabbed in phone fraud

EAST BRUNSWICK — A man who described himself as an electronics engineer has been arrested on charges stemming from the use of a "blue box," a gadget the size of a calculator that emits electronic signals that bypasses regular telephone billing equipment.

Tarkeshwar Singh, 50, of 18 1st Place, was freed on his own recognizance after he was arrested yesterday in a public phone booth at a Route 18 department store. Detective Donald Henschel reported.

Singh was charged with possession of a burglary tool, the "blue box," and theft of \$300 worth of services from

New Jersey Bell Telephone Co. police said.

Investigator James Witaneck of the phone company's security division in Newark, said the investigation had been in progress for several months. During that time, he said, Singh used the device for \$300 worth of phone calls to Japan and Hong Kong.

In addition to the "blue box," investigators confiscated a schematic design of the instrument which they said had been sent to Singh by an acquaintance in West Germany.

"These electronic devices are a continuing problem to the telephone company," Witaneck said.

THE POSTAL (DIS)SERVICE

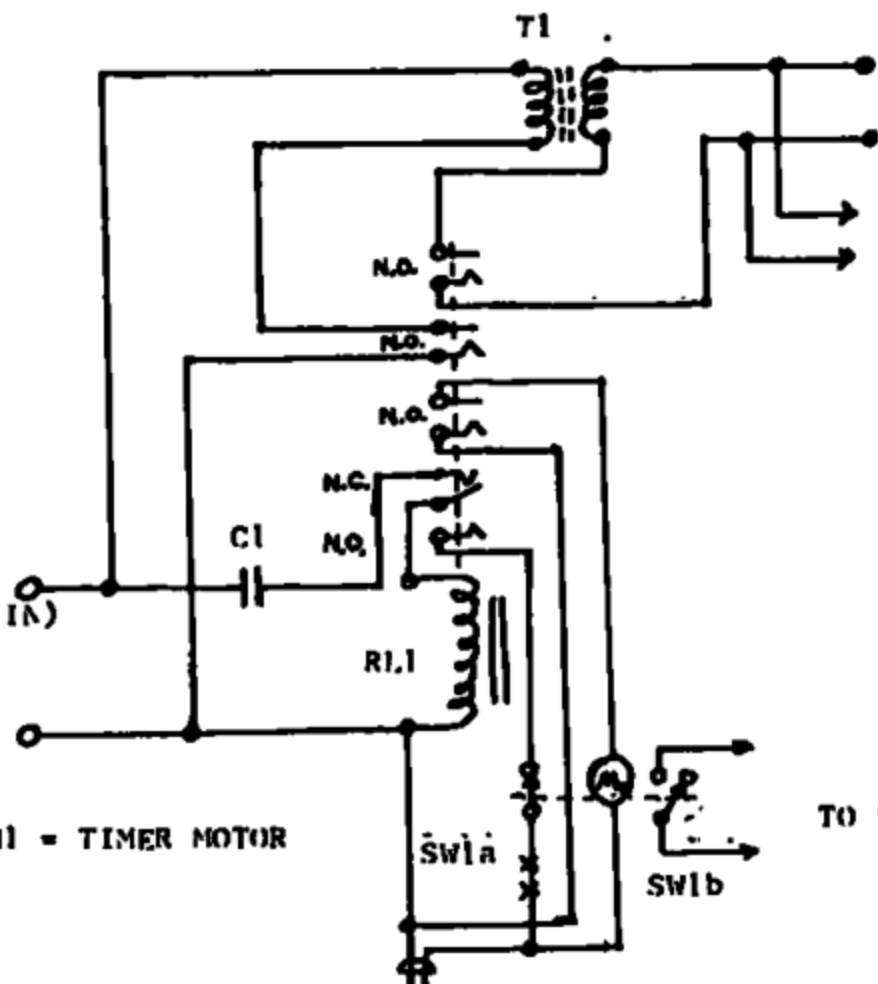
The Department of Agriculture recently completed a major survey, packed up its findings in some 1,500 boxes and mailed them off to its Des Moines, Iowa, processing center. While in the fumbling hands of the Postal Service, more than 600 of the parcels were damaged, but or delivered to the wrong address.

We trust that when the Agriculture Department lodged its complaint, it had the good sense to call rather than drop a letter in the mail.

TO PHONE LINE #1 (IN)



MI = TIMER MOTOR



DO IT YOURSELF CALL FORWARDING DEVICE

Calling his buff

LAS VEGAS (AP) — What do you say to a naked burglar?

That's what police were wondering at 5 a.m. Sunday when they arrested Karl Hunsaker, 38, of Las Vegas, as he was climbing down a ladder in the buff carrying household goods from an apartment.

Hunsaker was booked for investigation of burglary. Officers gave no reason as to why Hunsaker had no clothes on.

TO PHONE LINE #2 (OUT GOING)

TO "NAME CALLER" DIALING CONTACTS

Jail phone line busy

DELAND, Fla. — A defendant usually gets to make one free phone call, but for a few inmates at the Volusia County Jail that apparently wasn't enough. Using coin-operated telephones in the jail, at least six inmates made \$32,000 worth of illegal calls, according to Assistant State Attorney Horace Smith. The inmates charged the calls to fake credit cards or to telephone numbers of unsuspecting citizens in this central Florida city, he said. Three inmates have been found guilty of charges in connection with the telephone case, and three others are awaiting trial, Smith said.

TO "NAME CALLER" START SWITCH

Dear TAP:

TO 110 VAC

In response to several pleas from your pub, enclosed is some technical data on the Pacific Telephones in Pasadena.

On Hook: 45 VDC

Off Hook: 7.5 VDC @ 60 ma

Phone input res: 200 ohms

Ring: aprox 50 VAC (My cheap multi-meter doesn't read AC mils)

T1 (mic button) res: 600 ohms

U3 (ear piece) res: 20 ohms (leads feeding earpiece show 80 ohms across them)

Ringer coil res: aprox 3Kohms. Only one coil.

Ring Back # 6105-6: (Prefix)-1-(Prefix) gets a weird "ticky-tock" sound; (Prefix)-0002 gets a nice 1000 cps tone; (Prefix)-1118 gets a real loud tone; (Prefix)-0000 gets a central office recording, which includes the unlisted phone number for the office (in this case 576-6119);

What was supposed to be the verifying number (Prefix)-1111, gets the "Not in service" recording; (Prefix)-0003 gets the referral operator; (Prefix)-0119 is a private party's home fone;

I'm trying to come up with a design for a "Dial through Cheese Box" sort of a gimmick, but the best I've been able to come up with is enclosed, but it's not what I want. I could do it if we had T-T phone hereabouts, but we're stuck with impulse dials. Drat.

Any ideas?

DO-IT-YOURSELF CALL FORWARDING DEVICE

MATERIALS:

C1 = 1.0mfd @ 400 VDC

RL1 = 4P.DT Relay, 115 vac coil

T1 = Audio isolation xmformer, aprox 600 ohms impd, 100 to 200 ohms DC Res.

M1 = Timer Motor, 115 VAC 60 CPS

SW1a = First section of timer switch, set for aprox 3 min closed, 10 sec. open (due to circuit configuration, timer will self-index to "open" position of this switch).

SW1b = Second section of timer switch, set for minimum possible duration "on". Indexed to close after SW1a has come out of detent. This is the critical factor in choosing the type of timer. "on" duration must be less than time required for "name caller" to finish dialing.

ADDITIONAL ITEM REQUIRED, BUT NOT SHOWN:

1, ea. battery powered "Name Caller" dialing machine or equiv.

NOTE: Over-ride disconnect switch (Tone Sens. Relay?) may be connected at point x-x.

Michigan Bell Telephone Co., the giant corporate institution that touches all our lives and wallets, gets absolutely giddy whenever we reach for the telephone. Bell spends millions of dollars around the clock and calendar for advertising and public relations to persuade us to reach more often.

We are taught, however, obliquely, that we are disadvantaged unless we have telephones handy in every room of the house and office, or if we do not use them to facilitate every kind of communication. Shop by phone; sell aluminum siding by phone; solicit and collect money by phone. Telephone your mother, lover, great uncle and your entire high-school graduating class at least once a week, just for the kick of it. Get a separate line for the kids! Get a car phone! Give a phone to a poor person! And, do invest in one of those recording devices so you will never, never miss a call, even a wrong number. That way Michigan Bell will never miss collecting for the call. We will all live happily ever after.

One ringy-dingy . . .

All this, and more, is the implied message of Bell's advertising. I have no argument with it. I would rather write than phone, or receive a letter than a call. Written words are special to me. Spoken words transmitted by electronic devices may be special to other people. The absurdity that intrigues me is not in the advertising or even in the concept of the telephone as an extension of the human mouth and ear.

Here's the absurdity: In its ever-diligent determination to expand service, the telephone company has opened 33 new Phone Centers around the state during the past 18 months or so. These are retail stores, more or less, in which you can purchase telephones (Bell calls them "instruments") and also arrange for installation when necessary, straighten out billing problems and generally do your telephone business.

These places are designed as walk-in centers, however. Therefore — and here it comes — they are not included in telephone book listings. This is not an oversight. The Phone Centers have unlisted telephone numbers. This is what I call absurd, remembering everything Bell has said about how essential telephone communication is to life itself.

I know about this because a fellow named Jerry Moon (I think) telephoned me to tell me about it. He had seen one of these Phone Centers near Telegraph and 13 Mile and wanted to dial it up to ask telephonic questions.

Two ringy-dingies . . .

"I couldn't find a listing," he said, "so I dialed the information operator. It rang 20 times and I hung up. Then, I dialed Bell headquarters, and someone there told me the number is unlisted. The person said Bell doesn't want its people bothered by phone calls. I swear to you that's what I was told."

I believe him. I confirmed it with a telephone company spokesman. He said, "The Phone Centers do have telephones, but we discourage telephone contacts. It's supposed to be a face-to-face operation. You know, a retail outlet to shop for phones."

He told me a lot of other things. He said Bell workers at the Phone Centers do not have access to central records and are not really set up to help with billing problems or repair problems. He said they have to refer all those things to other departments, which they will do, that that's a nuisance for them and a hold-up for customers. He said the main job of the Phone Centers is to sell phones and to arrange for service, and that having an unlisted number helps. All that translates to me as the same thing as, "Bell doesn't want its people bothered by phone calls."

I love it. Finding a giant absurdity is as exciting as finding the great pumpkin. Now, it's your turn.

Computer 'erases' phones

A malfunctioning computer board almost caused a total communications blackout yesterday at the Union County administrative complex in Elizabeth.

All 945 telephones at the county complex went dead shortly before 11:30 a.m. when a memory board in the telephone operations room burned out, according to James Delaney, director of central services.

Delaney said critical county opera-

tions, such as police and emergency communications, had been carried out over the county's radio system during the nearly four hours it took for the telephone company to restore service.

In the meantime, the county's work force either waited until the telephones were operative, or opted to "hoof it" between various floors or buildings in an effort to maintain communications until the system was repaired just after 3 p.m.

ABUSE OF REMOTE ACCESS SYSTEMS

John Petrie has a problem. Petrie (not his real name) is the communications manager for a medium size company in the Midwest. His company has installed a long distance control system to monitor usage and get better utilization of long distance facilities. Because the company has a large number of people traveling, remote access to the company's long distance facilities was installed to reduce the number of credit card calls. A series of inward WATS lines are connected to the long distance control system at headquarters. When traveling, company representatives can simply dial an "800" number and then their personal authorization code to get access to the company's long distance facilities including toll and outward WATS.

The remote access system seemed to be working great. Credit card calls had been all but eliminated and the overall cost had been reduced. Then about six months ago, Petrie was in the midst of doing the detailed monthly billing of calls to station users when he noticed that one person had been making a large number of 800 number calls via the remote access. Petrie thought to himself, "This guy's got to be a stupid fool to dial our 800 number to place a free 800 number call!" When questioned about the calls, the man denied making any remote access calls at all that month.

Totally confused at this point, Petrie called several of the 800 numbers listed on the billing report. In every case, when the call was answered, the familiar tones indicating entrance to a remote access system were heard. A phone freak clearly was at work!

Petrie immediately changed all codes, pauses and methods of gaining access to the company's system. That night, the mysterious caller tried 60 times before he finally figured out the new procedures and codes. Petrie made another major change, but the caller cracked that in about 20 tries, and then placed a call to Germany. Petrie removed international dialing from the system and called the telephone company security department.

Meanwhile, he decided to have some fun by calling the 800 numbers on the billing report, contacting each company's switchboard operator and asking to be connected to the communications manager. According to Petrie, "The moments of silence were deafening when I told these managers how I had reached them."

After about a week, the telephone company security people showed up and after reviewing the documentation were amazed. They traced all of the called numbers, and came up with nothing but remote access numbers, "meet-me" conference numbers and services such as Time and Temperature in upstate Michigan. They did their best to trace calls back to the originating number, and came up with calls from California out of another company's remote access system.

Petrie says that to date his company has been hit with about 6,000 fraudulent calls, which cost about \$10 an hour. "Even with all this," he says, "I don't feel we look too bad compared to companies I know who have been hit for in excess of \$2,500 a month on international calls alone. He seems to take great delight in calling Hertz Rent-A-Car on Guam."

John Petrie's problem is not unique. An informal survey by BCR reveals that a number of large companies, although by no means all, have had some type of a problem with unauthorized use of remote access facilities. Indeed, at least one large consulting firm has been investigating this problem for several clients.

The difficulty in getting access to a company's long distance facilities via remote access varies considerably. The system used by Petrie's company is one of the more difficult to crack in that it requires knowing the proper inward WATS number plus a valid authorization code. The system used in AT&T's Dimension PBX may be less secure in that there is one common access code for everyone. In some systems, no access or authorization code at all is required. Simply dialing the special local or inward WATS number gives the caller immediate access to the long distance facilities. The communications

Wrong line, indeed

TEMPLE, Texas (AP) — If you're one of those people who always seem to be caught in the slowest-moving line at the bank, you might understand the predicament a would-be robber found himself in recently.

The fellow stepped up to a teller at the First National Bank of Temple and demanded that she fill his sack with money.

"Give me the money, this is a stickup," the unarmed man told Claudine Holder.

Holder barely glanced at the canvas bag on her counter. Instead, the teller, whom bank vice president Sam Farrow described as "feisty and very quick-witted," informed the man that he was in the wrong line.

She directed him to stand in a line across the lobby, and while he waited meekly for service, she called police.

The suspect was arrested and charged with attempted bank robbery.

Army wants to find long-distance cheats

TACOMA — The Army wants to reach out and touch a few individuals who like to make illegal, long-distance calls at Fort Lewis.

One soldier phoned a number in the Dominican Republic and charged the \$1,288 call to the base.

Another has been calling from a pay telephone on Fort Lewis to a pay phone outside the Howard Johnson Restaurant in Trenton, N.J., and has been charging those calls to a number on the post. Worse yet, says Bill Wood, a base spokesman, return calls from Trenton are billed to the same number.

Ordinarily, long-distance bills going through the Fort Lewis communications center average \$2,000 a month. In May, the bills came to \$4,500. "Probably half or more of them are fraudulent," Wood says, "but we are checking and we will find these people."

Those making such calls could be imprisoned for five years for the offense.

manager of one large company says that his organization once used inward WATS to access long distance facilities through a Centrex system without any restriction. A caller simply dialed "9" and got access to the world. In one month there was \$5,000 to \$6,000 in unauthorized calls to destinations such as Israel, Hong Kong and Portugal. Belatedly, the company changed the system to restrict remote access calls to the company's tie line network.

A consultant who has studied the problem believes that most abuse of remote access to long distance facilities involves insiders or other persons closely associated with the company. Often, it is a customer or a supplier who finds out how to use the remote access. Sometimes it is just the difficulty in keeping authorization codes from becoming common knowledge within an organization. One company the consultant recalls was using MCI Execunet service, and the authorization code was supposed to be known by only a small group of persons. Eventually, it became known by a very large group. "I don't know how much security you can really put into it," the consultant says, "because once you tell the secretaries and they have to write memos to someone else, it is very hard to clamp down on it."

One of the country's largest manufacturing firms uses an operator-controlled system in which someone calling from outside wanting to use the long distance facilities must give the operator a four character code. The communications manager told BCR that while abuse is "not a significant problem for us, we know that there are people using the network who are not authorized to do it. Some of them are retirees from the company who have been around for a while and know the score. With 10,000 authorization codes, it is not too difficult to find a good one."

It appears that most cases of abuse are the result of people wanting to make free telephone calls. But there also seems to be an element of pranksterism involved. One company in the East, located near a large university, found they had a lot of outsiders accessing their telephone system. Suspecting university students, they got permission to install a call data recorder on the main university Centrex system. The data they collected confirmed that the students were, indeed, living up to their reputation for technical wizardry. They had not only found out how to access the company's long distance facilities, but its computer system as well. Fortunately, they

had not yet found out how to obtain or manipulate data in the computer.

John Petrie says one of the pranksters' tricks "is to place a call to Company A's remote access. From Company A's system, they then call Company B's remote access; then call from Company B to Company C; then call from Company C back again to Company A and finally to a non-releasing Time and Temperature number that, of course, will never hang up. By doing this on a Friday evening (none of the companies being aware of it until Monday morning), they can tie up entire systems for many hours of overtime charges."

How easy is it to find a remote access number? If you have some association with a company that has one or with the telephone company, the answer is probably: not too hard. But if you have no inside information, the difficulty is much greater.

To find out how hard it might be for an outsider, we decided to become a phone freak, and try to find an inward WATS line connected to a remote access. AT&T says that there are about 40,000 interstate inward WATS lines, of which about one-half have unlisted numbers. Presumably, a small percentage of these unlisted numbers are for remote access. Our experience suggests they are not easy to find.

Knowing nothing about how the telephone company assigns inwards WATS numbers, we began by consulting a readily available directory of listed 800 numbers to see if there was any pattern to how numbers are assigned. Our assumption was that unlisted numbers would follow the same pattern as listed numbers, an assumption that seems to be true.

It appears from the directory that 800 numbers do have some pattern: that the digits in the exchange code vary with the geographical area. WATS lines in New York, for example, have exchange codes that begin with a different digit than WATS lines in California. (We deduced the location from the fact that the listing said that the number was good anywhere except New York or except California.)

Knowing that a lot of company headquarters are located in New York, we selected some exchange codes that appear to be used very frequently in New York. We dialed these codes with varying combinations of the last four digits. After getting three answers and six recorded announcements saying the number was no good in the first ten tries (one number did not answer), we further analyzed the digits and dialed 30 good numbers out of the next 40. None of these numbers, however, was connected to remote access. After these 50 unsuccessful attempts, we got bored and gave up, deciding we were not cut out to be a phone freak. But had we more perseverance or an automatic dialer, perhaps eventually we would have found a remote access system. Of course, even if we had, we would be only half way home if the system required an authorization code.

It is this difficulty in getting through the security precautions that makes most observers believe abuse of remote access results generally from inside information. For the user being hit, this distinction might seem academic but it does suggest that a company can cut its losses substantially by concentrating on more internal security. The following are some effective measures:

1. Require a proper authorization code in addition to the access number.
2. Assign remote access authorization codes to a minimum number of people.
3. Provide enough digits in the authorization code so that you need assign only a small percentage of the maximum number of combinations.
4. Change authorization codes frequently.
5. When someone with a code leaves the company, retire the code.
6. If possible, install a system which tells you if a series of invalid codes has been dialed in.
7. Never give information on remote access to someone you do not know. A while back, an individual posing as an Action Communications Systems employee was calling WATSBOX users and asking for remote access numbers and codes, ostensibly to update Action's records. The caller was not from Action.

These precautions should minimize abuse of remote access, but they will not eliminate it. Ask John Petrie. He knows.

How To Cheat Your Ass Off In Skool

"I HAVE ONLY LEARNED BY COPYING"
—Mao Tse-tung

MAKING IT: Nice people just don't cheat. This is a fact of life. If you do cheat, you are most likely a rotten no good stinker with commie friends, dirty underwear and a host of social diseases. The Revolutionary 3 Stooges try to discourage this type of behavior. It is both tacky and unsophisticated. We suggest that instead, you follow the advice of our friends from Take Over, in Madison Wis., by just forgetting the entire mess. Fuck Skool! Forget cheating. Print up your own degree instead and get on with living.

(1) Borrow a friend's diploma, put your name on it and make a copy suitable for framing. You can take the signature from the old diploma, and get a facsimile when the new President is named—he will probably have his signature in the papers or on all kinds of documents.

(2) If you have a Gemini friend, get the friend's transcript and put your name at the top—if the friend has a degree. Again, make a copy.

Or if you have been here one semester—and don't rush, you have 4 years to graduate the Take Over way—you can get your own transcript and simply fill it in with courses it might have been nice to take. Reduce-verox your work to fit the form.

Consolidated Company in Chicago, a Saturn (discreet) firm will sell you a seal that works like a notary's seal for the transcript—you must emboss your list of courses and grades to give it that official look. You design the embossing seal yourself; put your birth sign in the center if you like, some Louis on the outside, with the words "University of Wisconsin." For Latin phrases we suggest a little joking, such as PECUNIA LOQUITUR.

FAKING IT: It was the morning after. After that is, dragging myself from the gutter in front of the Moonlight Bar to the back seat of my car. A bristly black hairy tarantula ran screaming from my mouth. Unknown substances mingled with cigarette butts in my hair. I had a mid-term exam in ancient Chinese history in 2 hours. You could say that I was unprepared. I asked myself, "What would Mao tse-fly do in a case like this?" But the Red Guards were nowhere in sight. I was on my own. I entered the class, paused and slowly labeled my blue book #2. I took my time writing a single grandiloquent concluding paragraph and handed it in. The professor later apologized for losing my first blue book and gave me a B. A cheat must always be resourceful. 1) Change the answers on graded tests. Bring them back to the prof and say, "Hey, I had this answer right." 2) Carry in completed blue books to the exam. 3) At the end of the quarter professors leave graded tests and term papers in the halls for their students. Take the best ones and save them for future use. 4) Keep all tests and papers to use again and again, use your friends' and visit fraternity files. 5) Remember to never put down what you plagiarized from as a source. Use master theses from other colleges, the papers kept by departments at other colleges for the "serious researcher" and obscure books from other libraries. 6) Despite propaganda, term paper companies are OK.

TAKING IT: I know of one student who walked into the school print shop as exams were being run off, sat down on a inked gally and walked off with a set of tests on his pants. 1) Bribe or get friends who can get tests, such as janitors and print shop workers. 2) Go through waste paper cans for copies.

CRIBING IT: What I have come to call the "Ethiopian Shuffle" was given to me by a foreign exchange student and has proven to be one of the best crib notes in the business. Taking a long narrow strip of paper that is folded like an accordion into a tiny book, you are able to write 10 times the amount of info that a normal crib sheet holds. It is then manipulated with thumb and forefinger. 1) Magic shops have special pencils which write invisible notes that can be seen with special glasses. 2) Intelligence is transmitted to several cheaters through an elaborate signal system. Pen point up is true, down is false. In multiple choice, fingers at chin level mean number of question—at waist level, number of answer. 3) Put cribs on the seat near your crotch. Open your legs to see it, close them to hide it. 4) Transistorized tape recorders can be camouflaged as hearing aids. 5) Be imaginative. Hide notes everywhere. On skin and fingernails. As scrolls in objects such as watches and pens. On kleenex, gum & cigarettes. Write on the sole of your shoe near the heel for easy reading when crossing legs. On tape in the folds of clothes and behind sheer nylon. *VIVA LARRY, CURLY MOE*

REVOLUTIONARY 3 STOOGES BARRAGE Box 166, NIGHT BRASSIA, DAYTON OH 45409

A TRANSIT worker who took it upon himself to tackle the TA's \$1-million-a-year problem with slug tokens has come up with an ingenious \$1 solution.

Thomas Costa, a 40-year-old turnstile foreman from Astoria, invented what he calls a "roll pin" device at home.

"We were having a problem at the Greenpoint Av. station where thin steel slugs were showing up regularly in token clerk buckets," he explained.

"I came up with it for

this one particular slug, but when I brought it in we found out it worked on all kinds."

The device works by measuring the width of the phony coins and dropping through those coins that don't fit the dimensions of legitimate tokens.

Forty copies of Costa's home invention were tested in several high-volume stations in Manhattan with "excellent success," a Transit Authority spokesman said yesterday.

This week, the TA ordered devices for installa-

tion in every turnstile in the system.

Slugs and foreign coins, which have plagued the subway system since it was opened nearly 80 years ago, have been used at epidemic proportions since the fare was hiked to 75 cents last month.

Nearly 30,000 fare beaters have dropped phony tokens into turnstiles every week since the fare increase, according to transit police.

A peak in slug use was reached in 1978 when 100,

000 phony tokens a week were being used.

Costa submitted his invention to the TA through its employee suggestion program, which means he gives up all patent rights to the device.

"The TA couldn't pay [employees] for all the stuff they've come up with," Costa laughed.

"What did he get for developing this thing?" Costa's boss, Joe Spencer, was asked.

"I kissed him twice," Spencer said.

New electronic "watchdogs" are making it increasingly difficult to fool Ma Bell.

The watchdogs are computer monitoring systems that have been set up to fight telephone toll fraud, which cost New Jersey Bell Telephone Co. millions of dollars last year through phony credit card numbers, fraudulent third-party billing and the use of electronic devices to bypass automatic billing equipment.

The loss due to electronic fraud only can be guessed at, since the devices work by circumventing company billing, but Bell spokesman Ted Spencer said the company had lost \$2.3 million through more conventional fraud schemes in 1980. The costs eventually are passed on to customers.

Company officials say telephone bill cheaters come from all segments of society, including college students, immigrants, middle-class suburbanites, businessmen and the poor.

For example, a 70-year-old Paterson woman recently was caught charging more than \$7,000 in overseas telephone calls to Greece using a "blue box," a device that emits tones reproducing the signals that guide telephone switching equipment.

Last week, an Israeli couple was charged with making calls to Israel with a blue box from pay telephones throughout Union and Middlesex counties.

After an investigation by the telephone company, a computer analyst making \$45,000 a year was charged with making fraudulent credit card calls on his lunch hour to Iran.

John T. Cox, Bell's district staff manager of security and investigations, said the detection systems for illegal electronic devices were getting better all the time.

"If you're using a blue box on a regular basis in New Jersey, you're going to be caught," he said flatly. "I can almost guarantee it."

Escorting a visitor through a seldom-seen computer room at Bell, Cox pointed out teletype monitors that can pick up the use of the device and immediately tell investigators where a call is being made from, so that cheaters frequently are arrested by local police while still on the phone.

Those found guilty of using a blue box can be fined, jailed and forced to make restitution.

A blue box is nothing more than a tone generator that gives its user access to the telephone company's long-distance lines by fooling automatic equipment. Users generally dial an 800 toll-free number and send a pulse that allows them to dial anywhere in the world without the call registering as a toll call.

The device was named for the color of the first boxes sold through underground publications, but they have grown in sophistication. Cox displayed several confiscated boxes built into small, handheld calculators and boxes the size of a cigarette pack. A young electronics engineer from Verona was arrested two years ago with a blue box he built directly into his telephone.

"The devices sell for up to \$300, but it's not worth it," Cox commented.

Bell prosecutes every blue box case it uncovers and works with police departments to move quickly in catching users. Because the blue boxes show no record of calls, Bell has run across cases of criminals involved in drugs and prostitution using the devices.

Cox said the use of blue boxes was falling off, explaining, "People are realizing they're going to get caught."

Since January, Bell investigators have come across 32 cases that have resulted in 12 arrests and 11 convictions.

Computer monitoring equipment also can pick out the use of other devices, such as black boxes, which avoid charges for incoming calls to a phone, and red boxes, which generate the sound of coins dropping in a pay telephone. Cox said new billing control systems soon would eliminate the electronic boxes.

Of more concern is non-electronic fraud, which Cox said was growing nationwide. It can range from charging a long-distance phone call to a stranger, to using a stolen credit card, but computers also are being put to use here.

Bell plans to introduce a special billing system that will need personal codes to operate, similar to autotellers being used by banks. Customers also will be able to stop anyone from billing a call to their number with an automatic computer block that signals an operator not to accept such calls.

However, it is impossible to stop all fraud. Cox pointed out, "The people who are perpetrating the frauds know our systems."



"Well, let's get busy. Where there's a tax law, there are loopholes!"